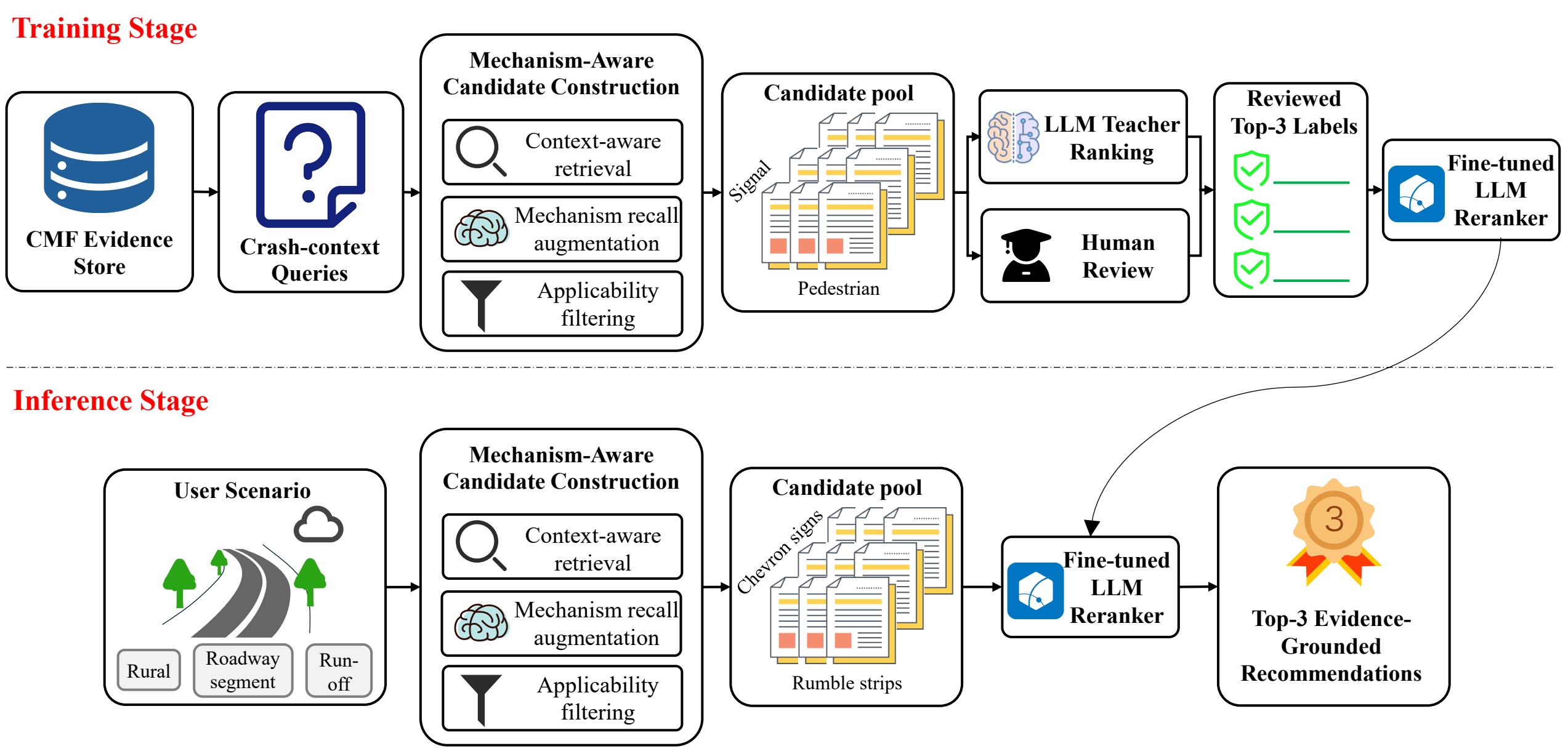

Training Stage
CMF Evidence Store
Crash-context Queries
Mechanism-Aware Candidate Construction
Context-aware retrieval
Mechanism recall augmentation
Applicability filtering
Candidate pool
Signal
Pedestrian
LLM Teacher Ranking
Human Review
Reviewed Top-3 Labels
Fine-tuned LLM Reranker
Inference Stage
User Scenario
Rural
Roadway segment
Run-off
Mechanism-Aware Candidate Construction
Context-aware retrieval
Mechanism recall augmentation
Applicability filtering
Candidate pool
Chevron signs
Rumble strips
Fine-tuned LLM Reranker
3
Top-3 Evidence-Grounded Recommendations